

[OTA] WBS for over-the-air updates

#	Work Package / Epic	Primary Component	Key Deliverables / Outcomes	Est. time (Days)
1	OTA System Foundations & Shared Infrastructure	Master + Agent	CMake structure, logging, config parsing, DisruptSDK wrapper, common error domain, atomic persistence utils	0.5
2	Master Startup, Registration & Vehicle Lifecycle	OTA Master	Config load, agent registration (late join), driving cycle events, basic policy engine	1
3	Resumable & Robust Download Subsystem	OTA Master	Persistent download state, HTTP range resume, pause/resume on power cycles, bandwidth throttling	0.5
4	New Version & Superseding Handling	OTA Master	Detect newer versions, supersede/archive old downloads, optional parallel version support	0.5
5	Post-Download Package Validation & Security	OTA Master	Size, CRC32, SHA-256, ECDSA P-256 signature verification pipeline	2
6	Manifest-based Compatibility & Targeting Engine	OTA Master	Parse manifest → per-agent eligibility (hw/sw/region/OEM rules)	0.5
7	Package Transfer & Acknowledgement to Agents	OTA Master ↔ Agent	Reliable file transfer, progress tracking, retry, negative acknowledgement handling	1
8	Safe-State Detection & Activation Orchestration	OTA Master	Vehicle safe-state collection → activation authorization → command dispatch	0.5

9	Post-Activation Health Monitoring & Stability Validation	Master + Agent	Multi-cycle error aggregation, threshold-based rollback decision	0.5
10	Master-initiated Rollback Coordination	OTA Master	Send rollback command, timeout handling, campaign state transitions after rollback	1
11	OTA Agent Core & Master Communication	OTA Agent	Agent startup, registration, heartbeat, command reception, status reporting	0.5
12	Software Artifact Passive Installation	OTA Agent	Extract to /opt/{pkg}/{ver}, handle /libs and /res, create/update newer symlink	0.5
13	Firmware Artifact Passive Installation (Staging)	OTA Agent	Stage to inactive slot/partition, partition health check, prepare rdinit if required	1
14	Software Artifact Activation & Minimal Downtime	OTA Agent	Switch symlink (newer → current), restart service/process, supervision integration	0.5
15	Firmware Artifact Activation (Controlled Reset)	OTA Agent + Bootloader	Coordinate reset in safe state, trigger bootloader slot switch, monitor boot result	2
16	Local Rollback Execution & Immediate Recovery	OTA Agent	On install/activate failure → revert symlink/slot, notify master, self-healing attempts	0.5
17	Post-Activation Local Health Monitoring & Reporting	OTA Agent	Crash/error detection, degraded mode signals, periodic health reports to master	1
18	Flash Bootloader Interface Definition (Abstract Layer)	Shared	Define BootloaderInterface abstract class (GetCurrentSlot, SetActiveSlot, MarkSlotAsBad, ValidateStagedFirmware...)	1
19	NVIDIA Jetson Orin Bootloader	OTA Agent	nvbootctrl integration, UEFI capsule support (if needed), rootfs/bootloader A/B switching, failure	2

	Plugin Implementation	(Orin HPC)	marking	
20	Generic U-Boot Bootloader Plugin Implementation	OTA Agent (classic ECUs)	fw_setenv/fw_printenv, bootpart/upgrade_available env vars, bootcount-based rollback, partition switch	2
21	Bootloader Plugin Factory & Agent Integration	OTA Agent	Factory to select plugin (config or node_identifier), call interface methods during firmware flows	1
22	Boot-time Image Validation & Automatic Fallback	Flash Bootloader	Verify image at boot, detect failed boot, switch to known-good slot (Orin: scratch registers; U-Boot: env vars)	3
23	Agent ↔ Bootloader Communication Protocol	Agent + Bootloader	Status exchange, activation trigger, failure notification	1
24	Artifact Packaging Standard & Tooling	Packaging Team	Manifest schema, /libs /res /exe /rdinit structure, CLI packaging tool, offline signing & validation	1
25	Self-Update & Master Failover Safety	Master + Agent (HPC)	Safe self-update of HPC running master, temporary handover to secondary master during own firmware update	5
26	Full System Observability, Audit & Diagnostics	All components	Campaign status aggregation, per-node progress, persistent audit trail, diagnostic interfaces (DBUS/file/cloud)	2
27	Testing	All components	integration e2e testing with test server	3